

DIGITAL MUNSHI AI

Legal Tech Assistant for Pakistan — Data Preprocessing Report

1. PROBLEM STATEMENT

Everyday citizens in Pakistan struggle to navigate the legal system due to complex legal language, high costs of legal consultation, and lack of structured guidance. Digital Munshi AI aims to bridge this gap using AI. This task focuses on establishing a clean, structured, and machine learning-ready dataset of Pakistani legal cases/queries categorized into six primary classes: **Criminal, Family, Property, Labor, Tenant, and Consumer**. The model will classify text queries to route them to appropriate legal drafts and automated guidance.

2. DATASET SOURCE & JUSTIFICATION

Source: Simulated Pakistan Legal Case Corpus (constructed based on realistic legal templates, statutes, and High/Supreme Court contexts).

Justification: Actual Pakistani legal documents are highly unstructured, inconsistent, and lack unified category annotations for supervised classification. Simulating this dataset with designated proportions, missing values, duplicates, and outliers ensures a robust, reproducible, and verifiable preprocessing pipeline before scaling up to larger real-world scrapes.

3. DATASET OVERVIEW

- **Dimensions:** 1215 rows × 8 columns.
- **Features & Types:** *case_id* (str), *date* (str/datetime), *court_level* (categorical), *language* (categorical), *client_type* (categorical), *user_query* (text), *severity_score* (float), *primary_category* (target).
- **Missing Values:** *severity_score* (60 nulls), *client_type* (37 nulls).
- **Duplicates:** Found 15 duplicate records.
- **Initial Class Distribution:** Criminal: 428, Family: 325, Property: 245, Labor: 98, Tenant: 68, Consumer: 51

4. PREPROCESSING STEPS PERFORMED

- **Deduplication:** Removed all duplicate rows to prevent data leakage during model training.
- **Missing Value Treatment:** Handled missing numerical values (*severity_score*) using Median Imputation and missing categorical values (*client_type*) using Mode Imputation.
- **Outlier Treatment:** Identified outliers in *severity_score* using Interquartile Range (IQR) and capped them at the $1.5 \times \text{IQR}$ boundary thresholds.
- **Feature Scaling & Encoding:** One-Hot Encoded categorical attributes (*court_level*, *language*, *client_type*). Scaled *severity_score* using Z-score Standardization.
- **Text Vectorization:** Preprocessed raw user queries (lowercasing, special characters removal) and vectorized using TF-IDF (term frequency-inverse document frequency).
- **Class Imbalance Resolution:** Used **SMOTE-Tomek** (Synthetic Minority Over-sampling Technique + Tomek Links) to balance categories, rising minority classes to match majority proportions, resulting in a balanced 6-class dataset.

5. LIBRARIES, TOOLS & SOFTWARE USED

Python 3, **Pandas & NumPy** (data manipulation), **Scikit-learn** (TF-IDF, Scaling, One-Hot Encoding), **Imbalanced-learn** (SMOTE-Tomek), **Jupyter Notebook / VS Code** (interactive development), and **Git/GitHub**.

6. CHALLENGES & RESOLUTIONS

Challenge 1: Text Feature Vectorization combined with Tabular Features. To apply SMOTE-Tomek, we concatenated TF-IDF word vectors with scaled tabular columns. **Challenge 2: Severe class imbalance.** SMOTE-Tomek successfully synthesized realistic samples for the minority classes (Consumer and Labor) while cleaning noisy decision boundaries by removing Tomek links.